Assignment Title: Implementing Multi-Factor Authentication System in Java

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Assignment Description:

Multi-factor authentication (MFA) is a crucial security mechanism that adds an additional layer of protection by requiring users to verify their identity through multiple independent factors. In this assignment, you will implement a simplified MFA system in Java. Your program will combine password-based authentication with a secondary factor such as a one-time password (OTP) or a security question.

Your program will:

1. Allow users to register with a username, password, and a secondary factor (e.g., mobile number for OTP or a security question).
2. Authenticate users by validating their password and the secondary factor.
3. Demonstrate the importance of MFA by simulating scenarios where one factor is compromised.

Requirements:

1. **User Registration**:
    - Allow users to register by providing a username, password, and a secondary factor:
        - For OTP: Generate and store a simulated mobile number or email for OTP delivery.
        - For a security question: Prompt for a question and answer.
2. **Password Management**:
    - Hash and securely store passwords using a hashing algorithm (e.g., SHA-256) with salt.
3. **Secondary Factor Implementation**:
    - **Option 1 (OTP)**: Generate a random 6-digit OTP during login and simulate sending it to the user.

    - **Option 2 (Security Question)**: Prompt the user to answer their security question.

4. **Authentication**:
    - Verify the password and the secondary factor.
    - Deny access if either factor is incorrect, and provide appropriate error messages.
5. **Error Handling**:
    - Handle invalid usernames, incorrect passwords, and failed secondary factor authentication.
    - Lock accounts after three consecutive failed login attempts.
6. **Documentation and Testing**:
    - Include comments explaining the functionality of each component.
    - Provide test cases demonstrating successful and failed authentication scenarios.
7. **Optional Enhancements** (Extra Credit):
    - Implement email or SMS OTP delivery using third-party libraries.
    - Save user data (hashed passwords, secondary factors) to a file for persistence.

Deliverables:

1. **Java Source Code**:
    - Submit .java file(s), ensuring the code is clean, well-structured, and documented.
2. **Test Results**:
    - Provide a document summarizing at least 3 test cases with:
        - Input credentials and secondary factors.
        - Outcomes of login attempts.
3. **Readme**:

    - Include a README file in Word or PDF format explaining how to run your program, its dependencies, and any optional features.

4. **Optional Enhancements** (Extra Credit):
    - Demonstrate file-based user data storage or email/SMS OTP delivery with additional examples.

Submission Guidelines:

1. Submit your .java file(s) and test results via Blackboard.
2. Include a README file in Word or PDF format explaining how to run your program, its dependencies, and any optional features.

10-Point Rubric:

Criteria	Points	Description
Correct Password Management	2	Successfully hashes passwords using a secure algorithm and stores them securely.
Secondary Factor Implementation	1	Implements OTP generation and validation or a security question/answer mechanism.
User Registration Functionality	1	Allows users to register with username, password, and a secondary factor.
Authentication Functionality	1	Correctly authenticates users by verifying both password and the secondary factor.
Error Handling	1	Provides meaningful error messages for incorrect inputs or failed authentication attempts.
Account Locking Mechanism	1	Locks accounts after a set number of failed login attempts.
Output Clarity	1	Clearly displays success or failure messages during registration and login attempts.
Documentation/Comments	1	Includes clear comments explaining major code sections and logic.
Test Cases and Results	1	Provides at least three test cases demonstrating the system's functionality.
Optional Enhancements	1	Implements file-based data storage or real OTP delivery via email/SMS. (Extra Credit)